# THE KUNNETH SPECTRAL SEQUENCE FOR FILTERED COMPLEXES

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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*Allowed standard prerequisites (editorial).*

- Existence of the filtered resolutions of the two complexes. Lemma \ref{lemma-resolution-filtered-complex}, cited at proof lines 15790-15791. Provides the resolutions whose tensor product is used.
- The spectral sequence of the total tensor product of two filtered complexes and its convergence criterion. Lemma \ref{lemma-kunneth-converges}, cited at proof lines 15795-15797. Supplies the spectral sequence and the boundedness and convergence statements.

*Source reference.* Original lemma 0H7Y: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 21473–21473. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

*Source reference.* Original lemma 0H7X: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 21472–21472. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

*Source reference.* Original lemma 0H7W: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 21471–21471. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

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*Date*: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

**Proposition 0.2.** *Let $R$ be a ring. Let $K^\bullet$ and $L^\bullet$ be filtered complexes of $R$-modules. Then there exists a filtered complex $T^\bullet$ representing $K^\bullet \otimes^{\mathbf{L}}_R L^\bullet$ in $D(R)$ such that the associated spectral sequence has $E_1$-page*

$$E_1^{p,q} = \bigoplus\nolimits_{i+j=p} H^{p+q}(gr^i K^\bullet \otimes^{\mathbf{L}}_R gr^j L^\bullet)$$

*If*

(1) *$F^i K^\bullet$ is acyclic for $i \gg 0$,*
(2) *$F^i K^\bullet \to K^\bullet$ is a quasi-isomorphism for $i \ll 0$,*
(3) *$F^j L^\bullet$ is acyclic for $j \gg 0$, and*
(4) *$F^j L^\bullet \to L^\bullet$ is a quasi-isomorphism for $j \ll 0$.*

*then the spectral sequence is bounded, the associated filtration on each $H^n(K^\bullet \otimes^{\mathbf{L}}_R L^\bullet)$ is finite and the spectral sequence convergences.*

*Proof.* Choose $P^\bullet \to K^\bullet$ and $Q^\bullet \to L^\bullet$ as in Lemma 0H7X. Then we use the spectral sequence for the filtered complex

$$T^\bullet = \mathrm{Tot}(P^\bullet \otimes_R Q^\bullet)$$

described in the text of this section and in Lemma 0H7W. □